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infDB: An Open-Source Data Ecosystem for Urban Energy Infrastructure Modelling and Planning

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Summary

`infDB - Infrastructure and Energy Database` is an open-source, containerized data infrastructure for managing and providing access to heterogeneous energy and infrastructure datasets used in urban and regional energy system studies. It bundles a PostgreSQL-based database with geospatial, time-series and graph extensions, standardized REST and OGC-compliant APIs, and a configurable import service that transforms raw public data into structured, version-controlled schemas.

By separating data ingestion, storage, and access from downstream analysis and modeling tools, infDB reduces data preprocessing effort and enables reproducible, transferable energy modeling workflows across regions and projects. Thus, it can be applied to diverse energy system modeling tasks, including district heating planning, electrical distribution network analysis, and urban energy demand estimation.

Statement of need

The transition to a climate-neutral energy system is a central pillar of energy policy, exemplified by Germany's aim for climate neutrality by 2045. New legislative frameworks, such as the requirement for municipal heat planning (KWP) and the requirement for grid expansion plans based on regional transition pathway scenarios defined in the German Energy Industry Act (EnWG §14d), demand that municipalities and Distribution System Operators (DSOs) process vast amounts of energy and infrastructure data (*BMWSB Kommunale wärmeplanung - BMWSB*, 1/19/2026; *EnWG §14d*, 1/19/2026).

However, the current landscape of energy data is fragmented. While the Open Data Strategy of the Federal German Government (*Data Strategy of the Federal German Government*, 1/27/2021) has increased data availability, this data is published by disparate authorities on different platforms in varying formats, spatial resolutions, and licensing structures. Consequently, energy modeling workflows often suffer from:

1. **Lack of Reproducibility:** Planning results are often one-off studies that are difficult to update or audit.
2. **High Pre-processing Effort:** Researchers spend disproportionate time acquiring raw data before analysis.
3. **Limited Workflow Transferability:** Data processing workflows often require substantial adaptation across regions due to differing data formats, interfaces, and conventions.

4. **Siloed Infrastructure:** DSOs and municipalities lack standardized tools to manage and share distribution network data efficiently.

`infDB` addresses these challenges by providing a reproducible, version-controlled, and automated ETL (Extract, Transform, Load) pipeline. It acts as a middleware between raw public data and high-level energy modeling tools, ensuring that planning data is transparent, traceable, and easily updatable. While `infDB` is developed in a German context addressing local regulatory requirements, the underlying architecture and methodology can be also applied to data across different regions and regulatory frameworks, making it applicable to municipal and regional energy planning efforts worldwide.

State of the field

Energy and infrastructure data management is an active field with several existing solutions:

- **Commercial Platforms:** Tools like **nPro**, **Solarea**, and **flexRM** offer robust analytics and user interfaces but are proprietary, limiting transparency and community extension.
- **Open Source Modeling Frameworks:** Tools like **City Energy Analyst (CEA)**, **EUReCA** and **OpenPlan** focus on modeling and optimization but typically assume preprocessed, structured input data provided externally.
- **Data Initiatives:** The **NEED project** (*NEED - Neue Daten Für Die Energiewende*, 9/01/2023) provides a decentralized data hub for synthetic energy data, and **DB4KWP** (*Vom Plan Zur Wende*, 1/19/2026) focuses on ontologies (OEO/OEKG) and naming conventions.

`infDB` fills the gap between available open data sources and existing energy modeling and planning solutions. While simulation tools like the City Energy Analyst focus on modeling, `infDB` provides the foundational data infrastructure that these tools require. Moreover, unlike static data repositories, `infDB` offers a dynamic, service-oriented platform that enables users to deploy local instances, continuously integrate fresh datasets, and seamlessly connect with both commercial and open-source downstream tools. Looking forward, `infDB` can complement ontology initiatives like DB4KWP by providing a technical implementation layer that transforms conceptual data standards into practical, operational systems.

Software Design

`infDB` is designed as a modular, containerized data infrastructure that separates data ingestion, storage, and access from downstream analysis and modeling. It follows a service-oriented architecture orchestrated via Docker Compose, allowing individual components to be deployed, configured, and combined depending on the requirements of a specific workflow. The system is conceptually divided into **Services**, which provide the foundational infrastructure, and **Tools**, which consume and process the data. This separation allows for high portability and enables users to activate only the components required for their specific use case.

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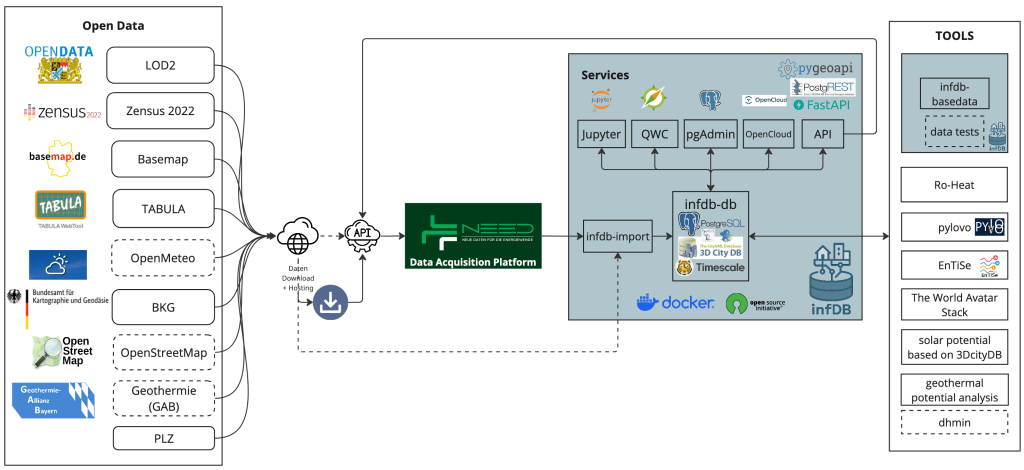

Figure 1: infDB - Data Sources, Services and Tools

The *Services* layer (depicted in the grey box in Figure 1) handles database operations, administration, data ingestion, and connectivity. These containerized services include:

- **infdb-import:** This service automates the ingestion of heterogeneous external data sources. It transforms raw external formats into structured schemas within the database. Users control this process via a simple YAML configuration file (`config-infdb-import.yml`), eliminating the need for custom ETL scripting.
- **infdb-db:** The central storage engine hosting a PostgreSQL database. It is pre-configured with essential extensions for energy modeling:
  - **PostGIS** for geospatial data.
  - **TimescaleDB** for time-series data.
  - **3D City DB** for (3D) semantic city models.
  - **pgRouting** for graph-based network analysis.
- **APIs and Data Access Services:** infDB exposes data exclusively through standardized interfaces rather than custom file formats. This includes SQL access to the database as well as RESTful and OGC-compliant APIs implemented using **FastAPI**, **PostgREST**, and **pygeoAPI**. These interfaces allow external tools to access data in a consistent manner while keeping the internal database structure encapsulated.
- **Administrative and Interactive Services:** Optional services such as **pgAdmin**, **Jupyter Notebook**, and a web-based **QGIS** client support administration, inspection, prototyping, and visualization of data stored in the database. These components are intended to lower the barrier to entry for users from different backgrounds (e.g., GIS specialists or researchers) but are not required for automated or headless workflows.
- **External Storage Integration:** An optional **OpenCloud** component allows integration with cloud-based storage solutions for handling large datasets. This component is not required for local deployments and can be omitted in minimal setups.

The *Tools* layer (depicted in the right box in the architecture diagram) consists of (external) software that interacts with the `infDB` Services to process data or generate insights. This modular approach allows users to chain different tools into custom workflows. Tools can be built upon following foundations:

- **Standardized Integration:** Tools interact with the core database exclusively through open interfaces (SQL or REST APIs), ensuring that the underlying data schema remains consistent regardless of the tool used.
- **pyinfdb:** To facilitate the development of custom tools, the platform provides the `pyinfdb` Python package. This library abstracts database connections, logging, and configuration management, allowing researchers to rapidly develop Python-based analysis scripts that

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integrate seamlessly with the infDB ecosystem.

- **Extensible Ecosystem:** Importantly, infDB does not prescribe specific modeling approaches, optimization methods, or planning workflows. Its role is limited to providing a stable and reproducible data infrastructure that can be reused across different analytical contexts. Therefore, users can integrate existing third-party simulation software or develop proprietary tools that plug into the `infDB` backend without modifying the core services.

Research Relevance

The research relevance of `infDB` lies in its role as a reusable data infrastructure that supports transparent and reproducible energy system analysis workflows. By separating data management from analysis logic, `infDB` contributes to several recurring methodological requirements in energy research:

- **Reproducibility:** Containerized deployment and configuration-based data ingestion allow complete data pipelines to be rerun and inspected. This enables studies to be reproduced or updated when new data becomes available, without reimplementing preprocessing steps.
- **Transferability:** By enforcing structured schemas and standardized access interfaces, infDB allows data processing and analysis workflows to be reused across regions and projects with minimal adaptation, reducing region-specific reimplementation effort.
- **Separation of Concerns:** infDB decouples data acquisition, storage, and access from modeling and analysis code. This allows researchers to develop and test analytical methods independently of changes in input data formats or sources.
- **Workflow Stability:** Stable database schemas and interfaces provide a consistent foundation for iterative research, supporting comparative studies and sensitivity analyses without requiring repeated adjustments to data handling logic.
- **Methodological Neutrality:** infDB does not prescribe modeling approaches, optimization methods, or policy assumptions. Its role is limited to providing structured and accessible data, allowing a wide range of analytical methods to be applied without constraint.

Applications

There are different applications to use `infDB` such as: * Calculating linear heat density by estimating building heat demands and distributing them along street segments to assess the financial feasibility of district heating as a basis for municipal heat planning (KWP) or district heating feasibility studies (BEW) * Generating synthetic data of existing low-voltage grid structures for grid planning to analyze impacts of new assets and their optimized management on grid reinforcement requirements.

The following section outlines the logical steps of the infdb-workflow, using the calculation of linear heat density for district heating planning as a practical example:

Step 1 – Data Integration: Raw, heterogeneous data sources (building registries, census data, network geometries, energy consumption records) are ingested through `infDB-import` and stored in the unified database, eliminating manual data collection and format conversion.

Step 2 – General tools: The toolset queries the database via REST or SQL APIs to extract raw data, which is then used to generate enriched base datasets such as the estimation of households per building or building-specific thermal properties. By replacing isolated, file-based GIS workflows with this interoperable framework, the system minimizes computational overhead and streamlining efforts.

Step 3 – Use case specific toolchain: The consistent data foundation provided by infDB enables the creation of reproducible synthetic data. In the context of KWP planners can apply

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domain-specific algorithms to compute building-level heat demands and spatially distribute them along street network segments. The resulting linear heat density maps directly inform feasibility assessments by identifying corridors where district heating networks are economically viable.

Step 4 – Iteration & Update: When new data becomes available (e.g., updated building stock, revised energy statistics), the entire pipeline can be re-executed without reimplementing preprocessing logic, ensuring studies remain current and reproducible.

In summary, the `infDB` ecosystem allows researchers and planners across various use cases to access the full spectrum of container-ingested data. This includes open data (e.g., building LOD2), enriched basedata (e.g., estimated households per building), and intermediate simulation results (e.g., heat demand). This also facilitates the seamless integration of downstream tools, such as energy system optimization models, with consistent data for both electrical and district heating grids.

AI usage disclosure

In the development of this work, GitHub Copilot, ChatGPT, and Gemini were used. Github Copilot assisted in code generation by code suggestions and completion tasks, while ChatGPT and Gemini assisted in drafting and refining textual content.

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